Laboratory Transition and Experimental Protocol for the Materialization Program: Seeded Crystalline Growth Tests

Companion operational manuscript

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Abstract

This manuscript operationalizes the lowest hard-materialization regime of the materialization program in apparatus-level language. It is a downstream laboratory bridge from the core theory manuscript and its computational companion rather than a proof document or a general experimental-discovery paper.

The companion defines a theory-to-laboratory translation map, apparatus-level target objects, measured observables, growth and persistence protocols, acceptance criteria, failure modes, escalation rules, and audit-trail requirements for seeded crystalline growth tests. Higher fabrication, device, and hierarchy claims remain out of scope.

The present pass is limited to the H0 regime together with a strict laboratory operationalization of seeded crystalline growth. It does not claim theorem closure, arbitrary-object fabrication, or universal extrapolation beyond the chosen target class.

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1 Protocol, Boundary, and Laboratory Status Policy

1.1 Purpose of the laboratory document

This companion translates the lowest materialization regime into experimental and apparatus-level language. Its purpose is to define a disciplined empirical bridge from theory and simulation to a physically programmable seeded-growth test sequence.

1.2 Boundary of the laboratory manuscript

The present manuscript is downstream to the core theory and the computational companion. It does not claim a proof of the theorem chain, general materialization of arbitrary objects, higher-hierarchy closure, or unrestricted transfer beyond the chosen seeded crystalline target class.

Shared boundary clause. This companion is downstream to the core theory manuscript; it is not theorem evidence for the core manuscript; and it operationalizes only the lowest regime through apparatus-level protocols and measured surrogates.

1.3 Laboratory status labels

The laboratory companion uses a dedicated status system:

- [LA] Laboratory Assumption.
- [LD] Laboratory Definition.
- [LP] Laboratory Protocol.
- [LM] Measured Observable.
- [LR] Laboratory Result.
- [LF] Laboratory Failure Mode.
- [LT] Laboratory Target Claim.

2 Translation Map from Theory to Laboratory Practice

2.1 Theory objects carried into the laboratory

The laboratory bridge translates theory objects into apparatus-level counterparts while preserving the boundary between theorem statements and experimental operationalization.

Theory object	**Laboratory counterpart**	**Measured or controlled channel**	**What is not claimed**
S_{seed}	Physical seed nucleus or template	Seed preparation, placement, and admissibility checks	No complete nucleation theory
$R_{\mathrm{pre}}^{(a)}$	Feedstock, vapor, melt, or solution reservoir	Precursor preparation and sufficiency control	No universal reservoir physics
u	Control schedule	Temperature, pressure, concentration, field, or boundary program	No canonical control trajectory
Γ_{grow}	Experimental growth sequence	Time-ordered run protocol and logged state transitions	No literal reconstruction of microscopic dynamics
Q_Λ	Reciprocal-order or diffraction metric	In situ or post-growth order measurement	No unique order observable
$\lvert D(X)\rvert_O$	Defect surrogate	Microscopy, linewidth, vacancy, or contamination proxy	No observer-independent defect norm
V_{occ}	Completion or fill fraction	Target-region occupancy estimate	No exact microscopic completion oracle
τ_O	Post-control hold time	Persistence check after control removal	No universal metastability timescale

2.2 What remains indirect or surrogate-level

The measured order, defect, completion, and persistence observables are laboratory surrogates for theory-level quantities. The present manuscript therefore treats them as disciplined empirical proxies rather than as literal replacements for the theory objects.

The current laboratory bridge is intentionally anchored to the imported P0 simulator line. It therefore treats the simulator not as proof, but as a pre-screening and crosswalk package for choosing target windows, verdict classes, and the minimum data products required in a first physical campaign.

2.3 Shared outcome ladder

The laboratory companion preserves the same high-level outcome ladder used across the companion package:

- fail before growth;
- growth fail;
- glass or amorphous capture;
- soft success; and
- hard success.

The laboratory document assigns these classes through apparatus-level measurements and post-control validation rather than through simulator-only or theorem-only criteria.

3 Laboratory Standing Assumptions

3.1 Material-system assumptions

Assumption 3.1 [LA]. Chosen material family. The active H0 laboratory release fixes the material family to copper homoepitaxy on Cu(111) witness substrates under ultra-high-vacuum thermal evaporation. Copper is the only active deposited species in the target window; alloying, reactive chemistry, and multi-species growth are excluded from this release package.

3.2 Geometry assumptions

The physical target window is fixed to a central $4\,\mathrm{mm} \times 4\,\mathrm{mm}$ square inside an $8\,\mathrm{mm} \times 8\,\mathrm{mm}$ Cu(111) witness coupon. The baseline admissible seed is a pre-annealed central seed patch of side length $1\,\mathrm{mm}$ aligned with the substrate axes so that completion, off-target spill, and parasitic islands can be normalized against a frozen geometric denominator.

3.3 Instrumental assumptions

The verdict ladder is defined only when the chamber provides: a base pressure no worse than $10^{-8}\,\mathrm{mbar}$, a controlled copper flux source, a quartz-crystal microbalance for dose logging, an in situ diffraction channel (RHEED or LEED), and post-growth microscopy by AFM or SEM. If any one of these channels is unavailable, the H0 verdict ladder is treated as underdetermined.

3.4 Boundary assumptions

The environment is assumed sufficiently isolated that contamination bursts, parasitic nucleation outside the target window, and temperature or flux drift remain auditable and suppressible. The post-control hold stage is always run with the copper shutter closed and no auxiliary field schedule active.

4 Experimental Target and Measured Observables

4.1 Laboratory target definition

Definition 4.1 [LD]. Laboratory target object. The laboratory target is the apparatus-level realization of Σ_O^{mono}: a seeded crystalline body with a chosen lattice class, a completion threshold, a defect tolerance, and a post-control hold requirement.

For the current lowest-regime bridge, the intended object class is the simplest seeded crystalline body compatible with a monatomic or effectively one-species growth regime. The first laboratory document therefore targets proof-of-regime seeded growth rather than arbitrary-shape fabrication or higher-level device functionality.

In the current release, that target is a contiguous Cu(111) patch grown outward from the central seed patch until the $4\,\mathrm{mm} \times 4\,\mathrm{mm}$ target window is filled to threshold while maintaining a single-domain diffraction signature and a bounded defect proxy.

4.2 Order observable

The order observable is the normalized diffraction-order score

$$q_{\mathrm{lab}} = \frac{I_{\mathrm{spec}} - I_{\mathrm{bg}}}{I_{\mathrm{ref}} - I_{\mathrm{bg}}},$$

where I_{spec} is the in situ streak or spot intensity assigned to the Cu(111) ordering channel, I_{bg} is the local background, and I_{ref} is the clean-reference Cu(111) intensity recorded during chamber qualification.

4.3 Defect observable

The defect observable is the weighted proxy

$$d_{\mathrm{lab}} = \phi_{\mathrm{void}} + 2\phi_{\mathrm{rough}} + 3\phi_{\mathrm{off}} + 4\phi_{\mathrm{cont}},$$

where ϕ_{void} is the void-area fraction in the target window, ϕ_{rough} is the fraction of the window whose roughness exceeds the run threshold, ϕ_{off} is the off-target deposition fraction, and ϕ_{cont} is the contamination or parasitic-island fraction estimated from post-growth microscopy.

4.4 Completion observable

The completion observable is the occupied-area fraction

$$v_{\mathrm{lab}} = \frac{A_{\mathrm{filled}}}{A_{\mathrm{target}}},$$

with A_{target} fixed to the central $4\,\mathrm{mm} \times 4\,\mathrm{mm}$ window and A_{filled} extracted from the fused in situ image stack and post-growth microscopy mask.

4.5 Persistence observable

Definition 4.2 [LM]. Post-control persistence observable. The persistence observable records whether the target remains admissible over the designated hold interval after the external control program has been reduced or removed.

For the current release, persistence is checked over a 30 min hold with copper flux off and substrate temperature stepped to a neutral hold band of 200±20°C. Hard success requires the target to remain above the order and completion thresholds and below the defect threshold throughout this interval.

5 Apparatus Architecture

5.1 Precursor subsystem

This subsystem uses 99.999% copper shot loaded into a thermal evaporation source, with deposition rate logged continuously by a quartz-crystal microbalance and frozen into the run sheet as the reservoir trace.

5.2 Seed or template subsystem

This subsystem uses a pre-annealed Cu(111) coupon with a central masked seed patch. The seed patch is admitted only if its pre-run diffraction image is single-domain to the resolution of the in situ order channel and if no parasitic islands are detected outside the masked seed region.

5.3 Chamber and environmental control

This subsystem fixes a UHV chamber, base pressure no worse than 10^{-8} mbar, substrate heating for growth and hold stages, and shutter control sufficient to separate active-growth and post-control windows cleanly.

5.4 Control subsystem

The control subsystem records the time-programmed schedule $u = (u_T, u_R, u_F, u_H)$ with baseline values $u_F = 0$, growth temperature 300 ± 20°C, copper flux 0.10 ± 0.02 monolayers per minute, and a post-growth hold at 200 ± 20°C for 30 min.

5.5 In situ diagnostics

This subsystem records QCM flux continuously, the order channel every 30 s, and a live occupancy estimate every 60 s during active growth.

5.6 Post-growth diagnostics

This subsystem records one diffraction image immediately after shutter closure and one AFM or SEM scan after cooldown, with an optional confirmatory LEED or XRD readout when the chamber workflow permits it.

6 Seed, Reservoir, and Control Preparation

6.1 Seed preparation protocol

The seed preparation protocol cleans and anneals the Cu(111) coupon, applies the central seed mask, grows the initial seed patch, and records a qualifying diffraction frame. A seed fails admissibility if the central patch is multi-domain, visibly contaminated, or misaligned relative to the witness-substrate axes.

6.2 Reservoir preparation protocol

The reservoir preparation protocol loads the copper source, verifies purity provenance, stabilizes the QCM trace, and freezes the nominal integrated dose before the run. The baseline dose target is set to 1.1 times the geometric fill estimate of the target window, with low-reservoir and high-reservoir controls fixed at 0.9 and 1.5 times this baseline dose.

6.3 Control-program specification

The control-program specification freezes the chamber qualification frame, the growth ramp, the active deposition interval, the shutter-close event, and the post-growth neutral hold. The baseline H0 line uses no auxiliary field program and treats temperature and integrated copper dose as the only active control families.

Control family	**Representative symbol**	**Operational meaning**
Thermal schedule	u_T	Time-programmed temperature envelope for seeded growth and post-control hold
Reservoir schedule	u_R	Feed or concentration schedule controlling precursor delivery
Field schedule	u_F	External field or boundary-conditioning program, when present
Hold schedule	u_H	Post-control relaxation and persistence-observation window

6.4 Pre-run admissibility checks

Protocol 6.1 [LP]. Pre-run admissibility checks. Before any run begins, the seed or template, the precursor reservoir, the chamber envelope, and the diagnostic channels must each pass a documented admissibility checklist.

7 Experimental Growth Protocol

7.1 Baseline seeded growth

The baseline seeded-growth run uses the frozen H0 package: Cu on Cu(111), dose ratio 1.1, growth temperature $300 \pm 20^{\circ}$C, and neutral hold 30 min at $200 \pm 20^{\circ}$C. This is the first candidate run for laboratory soft or hard success.

7.2 No-seed control

The no-seed control repeats the baseline run with the central seed step removed. Any ordered patch that forms is logged as an uncontrolled nucleation event and weakens the claim that the observed regime is genuinely seed-driven.

7.3 Low-reservoir control

The low-reservoir control uses the same geometry and thermal schedule as the baseline run but truncates the integrated copper dose to the 0.9 capacity-ratio analogue. It is the direct laboratory counterpart of the computational `FAIL_RESERVOIR` region.

7.4 High-reservoir control

The high-reservoir control raises the integrated dose to the 1.5 capacity-ratio analogue while keeping the same temperature program. It tests whether excess feed improves completion without causing parasitic islands or roughness runaway.

7.5 Strict-control run

The strict-control run narrows the growth temperature band to $300 \pm 10°\mathrm{C}$ and requires flux jitter below 5% of the nominal QCM rate. This is the laboratory best-case path toward hard success.

7.6 Relaxed-control run

The relaxed-control run widens the growth band to $300 \pm 30°\mathrm{C}$ and permits flux jitter up to 15% of the nominal rate. It is the deliberate stress case for defect-heavy growth, glass-like capture, or persistence collapse.

8 Measurement Protocol

8.1 In situ measurement sequence

This subsection defines when and how the order and completion surrogates are sampled during active growth.

At minimum, the laboratory line records growth time, active control settings, a running order surrogate, a running completion surrogate, and all flagged contamination or parasitic-nucleation events.

The baseline sampling cadence is one QCM stream, one diffraction frame every 30 s, and one occupancy update every 60 s.

8.2 Post-growth measurement sequence

This subsection defines which structural and defect observables are measured after growth termination.

The post-growth package includes one structure-sensitive readout, one defect-sensitive readout, one completion estimate, and a run-level contamination assessment so that the later verdict is auditable rather than narrative.

In the frozen H0 package, the minimum post-growth bundle is one final diffraction frame, one AFM or SEM mosaic covering the full target window, one roughness estimate, and one off-target deposition mask.

8.3 Post-control persistence sequence

This subsection defines the hold interval, the measurement schedule during that interval, and the criteria for declaring post-control persistence or collapse.

The persistence sequence is the direct laboratory analogue of the computational hard/soft split. It is therefore the decisive discriminator between control-supported success and post-control admissible success in the lowest regime.

The minimum persistence cadence is one diffraction frame at shutter closure, then at 10, 20, and 30 min into the hold stage, with a final microscopy snapshot after cooldown.

8.4 Uncertainty, repeatability, and data-quality rules

Threshold crossing is judged ambiguous whenever the measured value lies within ± 0.03 of the order threshold, ± 0.02 of the defect threshold, or ± 0.02 of the completion threshold. No verdict class is release-grade until it has been reproduced in at least three nominally identical runs.

9 Experimental Acceptance Criteria

9.1 No success

No success is recorded when the target object fails any one of the frozen thresholds $q_{\mathrm{lab}} \geq 0.65$, $v_{\mathrm{lab}} \geq 0.95$, or $d_{\mathrm{lab}} \leq 0.10$ under the active protocol.

9.2 Soft success

Soft success is recorded when the target object satisfies $q_{\mathrm{lab}} \geq 0.65$, $v_{\mathrm{lab}} \geq 0.95$, and $d_{\mathrm{lab}} \leq 0.10$ during active growth, but violates one of these bounds during the 30 min neutral hold.

9.3 Hard success

Target Claim 9.1 [LT]. Laboratory hard success criterion. *Hard success is recorded only when the order surrogate, defect surrogate, and completion surrogate satisfy the frozen thresholds* $q_{\mathrm{lab}} \geq 0.65$*,* $d_{\mathrm{lab}} \leq 0.10$*, and* $v_{\mathrm{lab}} \geq 0.95$ *during growth and continue to do so throughout the full* 30 min *neutral hold interval.*

10 Laboratory Failure Modes

10.1 No nucleation

Failure Mode 10.1 [LF]. No nucleation. *The precursor and control program fail to produce an admissible nucleus or seed-supported growth front.*

10.2 Parasitic nucleation

Parasitic nucleation occurs when off-target nuclei proliferate and undermine the intended seeded-growth path.

10.3 Polycrystalline divergence

Polycrystalline divergence occurs when multiple incompatible crystalline domains proliferate inside the intended target window.

10.4 Amorphous or glass capture

Amorphous or glass capture occurs when the run reaches a disordered trapped state instead of an ordered crystalline state.

10.5 Contamination-dominated growth

Contamination-dominated growth occurs when foreign occupancy or parasitic chemistry overwhelms the intended target regime.

10.6 Defect runaway

Defect runaway occurs when defect surrogates cease to remain inside the admissible measurement envelope during growth.

10.7 Post-control collapse

Post-control collapse occurs when an apparently valid target loses admissibility shortly after control reduction or removal.

10.8 Measurement ambiguity

Measurement ambiguity occurs when the laboratory readout channels do not separate the target verdict classes in a stable and reproducible way.

11 Relation to the Computational Companion

11.1 Parameter crosswalk

The laboratory companion maintains a parameter crosswalk between simulator knobs and apparatus control variables so that computational pre-screening and laboratory control remain comparable.

Simulator quantity	**Laboratory counterpart**	**Bridge meaning**
μ^*	Local process-compatibility envelope	Tolerance on local mismatch before a growth event is treated as operationally nonadmissible
δ^*	Incremental defect tolerance	Maximum tolerated defect jump per accepted growth stage
τ_O	Hold time after control removal	Laboratory persistence window distinguishing soft from hard success
δ_{rel}	Post-control collapse threshold	Maximum tolerated loss probability or decay indicator during the hold stage
v^*, q^*	Completion and order thresholds	Acceptance thresholds for measured fill fraction and order surrogate

11.2 Verdict crosswalk

The laboratory verdict ladder preserves the same high-level classes as the computational companion: failure before growth, growth failure, glass or amorphous capture, soft success, and hard success.

High-level class	Laboratory reading	Operational trigger
Fail before growth	No admissible start	No admissible nucleus, seed-supported front, or precursor sufficiency under the active preparation protocol
Growth fail	Incomplete or invalid growth	Growth proceeds but the measured order, defect, and completion surrogates never jointly cross their acceptance thresholds
Glass or amorphous capture	Trapped disordered outcome	The run ends in a disordered trapped structure distinct from the intended crystalline target
Soft success	Control-supported success	The measured target remains admissible only while active control remains in place
Hard success	Post-control admissible success	The measured target remains admissible over the designated hold interval after control removal

11.3 Pre-screening logic

The laboratory line identifies which runs require prior computational plausibility support before experimental execution is authorized.

In the present package, baseline seeded-growth runs and escalation to strict-control or high-reservoir conditions are gated by a computational pre-screening pass under the corresponding simulator-level target package and verdict ladder. The initial release authorizes laboratory execution only for simulator-side packages lying in the ordered-growth pocket $(\mu^*, \delta^*) \in \{(3.0, 0.3), (3.0, 0.5), (4.0, 0.3), (4.0, 0.5)\}$ with capacity ratio at least 1.1; laboratory hard-success claims are reserved for packages whose persistence analogue corresponds to the hard side of the imported P0 split.

12 Experimental Sequence and Escalation Policy

12.1 Proof-of-regime experiments

These experiments test whether the chosen apparatus can realize the lowest seeded-growth regime at all.

The initial proof-of-regime campaign mirrors the computational companion: one baseline seeded-growth run, one no-seed control, one low-reservoir control, and one relaxed-control run capable of exposing glass-like or defect-heavy capture.

The first release-grade campaign therefore consists of at least twelve runs: three baseline runs, three no-seed controls, three low-reservoir controls, and three relaxed-control runs.

12.2 Calibration experiments

These experiments refine the measured thresholds and the laboratory surrogate definitions for order, defect, completion, and persistence.

The calibration campaign uses a small family of manually curated positive, negative, and borderline physical states or runs so that the laboratory thresholds do not float free of a documented anchor library.

The initial anchor library is fixed to four physical families: clean filled Cu(111) patches as positive anchors, underfilled patches as completion-negative anchors, rough or parasitic-island-rich patches as defect-negative anchors, and borderline nearly complete patches as ambiguity anchors.

12.3 Stress tests

Stress tests deliberately probe the boundary between admissible growth and the documented laboratory failure modes.

12.4 Escalation rules

Escalation rules specify when the apparatus may proceed to harder conditions and when it must instead return to recalibration or computational pre-screening.

12.5 Stop rules

Stop rules specify when unresolved ambiguity, uncontrolled contamination, or repeated verdict instability blocks further experimental escalation.

At minimum, escalation must stop when: no admissible seed can be prepared reproducibly; contamination dominates the target window; the order and defect measurements cease to separate verdict classes; or post-control collapse remains unresolved under repeated nominally identical runs.

13 Safety, Reproducibility, and Data Governance

13.1 Safe operating envelope

The laboratory bridge defines the admissible operating envelope for the chosen apparatus, materials, and control schedules.

13.2 Repetition and reproducibility

The current release fixes three repetitions per condition, a maximum between-run spread of 0.05 in q_{lab}, 0.03 in d_{lab}, and 0.03 in v_{lab}, and blocks any public verdict claim when these spreads are exceeded.

13.3 Raw data schema

The laboratory line defines a normalized raw-data schema so that all measurements, control settings, and post-control observations remain auditable.

13.4 Audit trail and provenance

Every laboratory verdict is linked to an explicit run sheet, instrument record, threshold package, and raw-data snapshot.

The synchronized asset bundle for the current release is:

- `hardware/protocols/materialization_post_p0_growth_validation_protocol.md`;
- `hardware/bom/h0_cu111_baseline_bom.md`;
- `hardware/calibration/h0_cu111_threshold_package.md`;

- `hardware/calibration/h0_cu111_anchor_library.md`;
- `hardware/drawings/h0_cu111_geometry_spec.md`;
- `hardware/protocols/h0_cu111_run_sheet_template.md`.

14 What Is Not Claimed

The present laboratory companion does not claim a proof of the theory, arbitrary-object fabrication, higher-level closure, a universal process, or direct extrapolation beyond the chosen seeded crystalline target class.

A Materials and Instrument Table

This appendix lists the chosen material system, precursor forms, substrates or templates, and the instruments required for the active laboratory line.

Family	**Representative token**	**Role in the laboratory companion**
Material system	Cu on Cu(111)	Homoepitaxial effectively monatomic H0 target family
Precursor form	99.999% Cu shot in thermal evaporator	Frozen feedstock reservoir for baseline and control runs
Seed or template	Central $1\,\mathrm{mm} \times 1\,\mathrm{mm}$ Cu(111) seed patch	Physical nucleus supporting seed-driven outward growth
In situ order instrument	RHEED, with LEED when available	Diffraction channel used to estimate q_{lab}
Defect instrument	AFM or SEM mosaic	Channel used to estimate void fraction, roughness excess, and off-target spill
Persistence instrument	Repeated RHEED frames plus final microscopy	Hold-stage channel used to discriminate soft from hard success

B Measured Observable Definitions

This appendix maps each theory quantity to its laboratory surrogate, instrument channel, normalization rule, and uncertainty model.

Theory quantity	**Laboratory surrogate**	**Channel**	**Normalization question**
Q_Λ	Normalized Cu(111) diffraction score q_{lab}	RHEED or LEED	Clean-reference Cu(111) frame fixes the unit scale
$\lvert D(X)\rvert_O$	Weighted microscopy proxy d_{lab}	AFM or SEM	Void, roughness, off-target spill, and contamination fractions are aggregated with weights $(1, 2, 3, 4)$
V_{occ}	Window fill fraction v_{lab}	fused image mask	The denominator is the central $4\,\mathrm{mm} \times 4\,\mathrm{mm}$ target window
τ_O	Neutral-hold persistence record	repeated diffraction plus final microscopy	Hard success requires the full 30 min hold cadence to pass

C Experimental Run Sheets

This appendix contains the run-sheet template for baseline runs, controls, calibration runs, and stress tests.

Each run sheet should include the following blocks:

- target specification and lattice-class declaration;
- reservoir inventory and sufficiency check;
- seed or template preparation record;
- control schedule and intervention log;
- in situ observer-level diagnostics;
- post-growth diagnostic table;
- post-control persistence record; and
- final verdict assignment with trace archive identifier.

For the frozen H0 package, the control block must also record base pressure, nominal copper flux, integrated dose ratio, growth-temperature band, hold-temperature band, and timestamps for shutter-open and shutter-close events.

D Threshold Tables

This appendix freezes the active order, defect, completion, and persistence thresholds together with calibration notes.

Threshold family	**Representative token**	**Laboratory meaning**
Order threshold	$q^*_{\mathrm{lab}} = 0.65$	Minimum acceptable normalized Cu(111) diffraction score
Completion threshold	$v^*_{\mathrm{lab}} = 0.95$	Minimum acceptable fill fraction of the $4\,\mathrm{mm} \times 4\,\mathrm{mm}$ target window
Defect threshold	$\varepsilon_{\mathrm{lab}} = 0.10$	Maximum acceptable weighted defect proxy d_{lab}
Persistence threshold	$\tau^*_{\mathrm{hold}} = 30\,\mathrm{min}$	Required neutral-hold interval for laboratory hard success

E Raw Data Schema

This appendix records the normalized data fields required for every run, including control schedules, measurement timestamps, and post-control persistence records.

The minimum raw-data products of the first laboratory bridge are:

- target specification sheet;
- reservoir and seed inventory;
- growth trace;
- observer diagnostic table;

- persistence estimate; and
- final verdict record.

The frozen H0 line stores these as one run manifest, one QCM time series, one diffraction image stack, one post-growth microscopy bundle, one threshold-evaluation sheet, and one verdict record signed against the active threshold package.

F Safety and Escalation Checklist

This appendix provides the operator checklist for safe execution, escalation gating, and stop-rule enforcement.

1. Verify seed admissibility, precursor sufficiency, and contamination controls before every run.
2. Confirm that the active threshold package and control schedule are frozen before experimental execution.
3. Record all in situ and post-control measurements under the normalized run-sheet schema.
4. Block escalation when post-control collapse or measurement ambiguity remains unresolved.
5. Return to computational pre-screening when repeated verdict instability appears under nominally identical conditions.
6. Do not release a hard-success claim unless all three repetitions pass the full 30 min neutral hold with the frozen Cu/Cu(111) threshold package.